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Unfolding North American spring weather extremes along a scale ladder

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Springtime weather extremes have profound socioeconomic consequences across North America. While their dynamical linkages with synoptic- and mesoscale systems are well-studied, there is limited understanding of their variability on subseasonal and longer timescales. Here, we unfold the multi-layered dynamics of these extremes along a "scale ladder" built around canonical modes of subseasonal variability over North America in spring. The four leading subseasonal modes are characterized by localized geopotential anomalies over the North Pacific and northeastern North America, and two branches of zonally propagating Rossby wave packets. These modes modulate the activity of wet, dry, and wind extremes across North America, causing up to a twofold change in their occurrence frequencies. Over four decades, these modes have shown distinct "active" and "muted" periods, leading to decadal-scale changes in extremes. The multiscale perspective highlighted here provides a dynamics-based guideline for dissecting long-term changes of extremes and extracting potential sources of predictability across timescales.

Keywords North America, Weather extreme, Subseasonal variability

Weather extremes generate profound impacts on our societies, often incurring human suffering, environmental damage, and significant socioeconomic costs[1]. In recent decades, their occurrence frequency and/or intensity have been shown to increase across the globe, particularly in terms of temperature and hydrological extremes[2], leading to growing interest in a better understanding of weather extremes. North America stands out as one of the most prominent hotspots for extreme weather across the globe. Recent decades have witnessed severe precipitation- and temperature-related extremes in North America. For instance, catastrophic flooding events (e.g., 2019 Midwest floods), triggered by intense extratropical cyclones and/or tropical cyclones, produced adverse impacts on various sectors including agriculture and infrastructure resulting in billions of dollars in financial loss[3–6]. Persistent multi-year droughts on the west coast of the United States caused widespread damage to the local water cycles[7,8]. In April 2011, a series of intense tornadoes generated devastating consequences across the southern United States[9,10]. Moreover, North America has recently experienced unprecedented heat waves and cold air outbreaks, leading to significant loss of life and property[11–13].

As a transition season and an agriculturally-important growing season, boreal spring over North America is characterized by weather extremes of greater complexities and multi-layered dynamics in comparison to winter and summer due to the peak of the activity of extratropical storm tracks and mesoscale convective systems (MCSs)[14–20]. MCSs, fronts, and extratropical cyclones frequently occur downstream of the Rocky Mountains, explaining heavy precipitation, hails, and damaging winds in the central and southeastern United States[19,21,22]. Eastward-propagating extratropical cyclones or atmospheric rivers downstream of the North Pacific jet are responsible for the majority of the flood and mudslide hazards along the west coast of North America[23–26]. It suggests that extreme precipitation is often accompanied by multi-scale weather systems. Understanding the variability of these multi-scale systems is therefore crucial for predicting long-term changes of boreal spring extremes over North America.

Earlier studies have investigated the dynamical linkages between large-scale circulations and synoptic (or mesoscale) systems in North America, aiming to explain the variability of the associated extreme precipitation[16,20,27–29]. In particular, much attention has been paid to the impacts of tropical variability on springtime extremes. For instance, Rossby wave propagation from the tropics excited by the Madden–Julian Oscillation (MJO) can influence regional climate on subseasonal timescales[30]. These circulation anomalies create favorable conditions for convective systems (e.g., storms and tornadoes), thereby inducing regional

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precipitation extremes over North America[31–35]. The waveguides of MJO teleconnection are also modulated by the El Niño–Southern Oscillation (ENSO) as it modifies the seasonal background state[36]. This suggests the importance of a systematic understanding of multi-scale atmospheric variability in driving surface extremes. A "scale" gap, however, remains in extratropical subseasonal variability, which bridges planetary-scale circulation changes with local, meso- and micro-scale extreme weather[37–40]. This subseasonal variability (or subseasonal mode) is typically linked to atmospheric blockings[41–44] or weather regimes[45–49] persistent beyond the synoptic scale of several days and is responsible for severe and long-lasting surface extremes.

The subseasonal variability also by definition plays a crucial role in subseasonal-to-seasonal (S2S) prediction of weather extremes and thus in the early warning of hazards and the related emergency preparation and mitigation efforts[50,51]. The fact that the magnitude of subseasonal variability in spring precipitation over North America is comparable to its climatology (Fig. S1) further fuels the need for a comprehensive understanding of the springtime North American subseasonal variability and its relationship with various types of weather extremes. Building upon previous studies, here we comprehensively delineate the statistical and dynamical nature of spring subseasonal variability over North America. We then integrate this missing "scale" step into a conceptual "*scale ladder*" to effectively unfold the multi-layered dynamics of spring weather extremes, with a focus on understanding their spatial variations and activity changes across decadal timescales.

Results

Leading modes of springtime subseasonal variability over North America

The climatological distribution of subseasonal variability amplitude is first presented here in Fig. 1, quantified in terms of the standard deviation of the 10–30 day bandpass filtered geopotential height at 500 hPa (see Methods) over the Northern Hemisphere from boreal winter to summer (Fig. 1a-c). In winter, the spatial distribution of subseasonal variability shows two distinct maxima over the North Pacific and North Atlantic with a third peak over northeastern North America (or the western flank of the North Atlantic) (Fig. 1a)[52,53]. In summer, the overall variability is markedly weaker, and the centers of action move poleward compared to winter. Its two maxima extend zonally with a westward shift compared to winter (Fig. 1c). Spring, which is the transition period from winter to summer, exhibits a mixed spatial distribution of both seasons (Fig. 1b). The North Pacific variability is elongated from the northwest to the southeast along the northern flank of the North Pacific jet exit, whereas the North Atlantic variability is zonally distributed downstream of the jet. As in winter, a pronounced variability is also observed over northeastern North America. Between the marked variabilities over the North Pacific and northeastern North America, a local minimum of variability appears around the high-elevation regions along the Rocky Mountains.

Focusing on North America, the leading modes of springtime subseasonal variability are identified (Fig. 1d-g), which are statistically distinct[54] (Fig. S2), by performing an empirical orthogonal function (EOF; Methods) analysis of the bandpass filtered geopotential height over the Pacific-North American sector (170°–70°W; 20°–70°N; boxed in Fig. 1b). The first mode (Mode 1), which explains 21.7% of the total subseasonal variance in the region, is characterized by a localized height anomaly in the high-latitude North Pacific, with weaker circulations of opposite sign over its downstream and on its equatorial side (Fig. 1d). The presence of this mode is in line with the maximum subseasonal variability seen over the North Pacific (Fig. 1b). The second mode (Mode 2), which accounts for 18% of the total variance, exhibits a marked circulation north of the Great Lakes (Fig. 1e). Such a circulation pattern also reflects the local peak of climatological subseasonal variability over northeastern North America (Fig. 1b). Unlike the first two modes, which are largely localized in high latitudes, the third and fourth modes are characterized by wave trains spanning over the northern extratropics. Mode 3, which contributes 13.1% of the total variance, features a positive-negative-positive (or negative-positive-negative) tripole circulation anomaly with the three nodes respectively centered about the northeastern North Pacific, western North America, and eastern North America (Fig. 1f). Mode 4, explaining 8.5% of the total variance, features a southeastward propagating Rossby wave train from the Bering Sea to the southwestern United States with a shorter zonal scale compared to Mode 3 (Fig. 1g).

The temporal evolution of four subseasonal modes, represented by regressions of Z500 and its tendency onto principal component (PC) time series, is examined in Fig. 2. Mode 1 displays a negative anomaly over the North Pacific at lag − 3 days (Fig. 2a, e), followed by quasi-stationary development with a meridional dipole structure resembling the Western Pacific pattern[55] (Fig. S3a). It retrogrades westward and weakens by lag + 3 (Fig. 2i), consistent with low-frequency variability over the North Pacific[56,57]. Mode 2 features a positive height anomaly over the North Pacific with a downstream negative anomaly at lag − 3 (Fig. 2b). This resembles a reversed decaying phase of Mode 1, suggesting a possible connection between the two (Fig. S4). The negative anomaly of Mode 2 strengthens partly due to a Rossby wave train from the North Pacific (Fig. 2b), sharing some similarity with the development of the positive phase of the North Atlantic Oscillation[58] (Fig. S3b). The amplified anomaly tends to move westward at onset and decays due to wave packets from the North Pacific (Fig. 2f, j). In particular, the "anticyclonic phases" (positive geopotential anomaly at high latitudes) of Modes 1 and 2 are closely tied to atmospheric blocking[42,43]. Climatologically, blocking occurs at northern high latitudes about 10–12% of the time in boreal spring (Fig. S5a, c). When conditioned on the anticyclonic phases of Modes 1 and 2, the relative occurrence frequency of blockings increases up to three times compared to the climatological values (Fig. S5b, d), suggesting more persistent and intense climate extremes to the regions[42,43,59]. The temporal evolution of Mode 3 suggests an equatorial Pacific origin of the Rossby wave train (Fig. 2c). The height tendency is out of phase with the height anomaly, suggesting a propagating nature of Mode 3. It continues to propagate slowly eastward until lag + 3 days (Fig. 2g, k). Mode 4 resembles a wave train propagating downstream from the North Pacific (Fig. 2d, h, l). The propagation is in a more zonally-oriented direction compared to that of Mode 3. These modes are also similar to weather regimes identified over North America[60–63], suggesting a close dynamical relationship between the two.

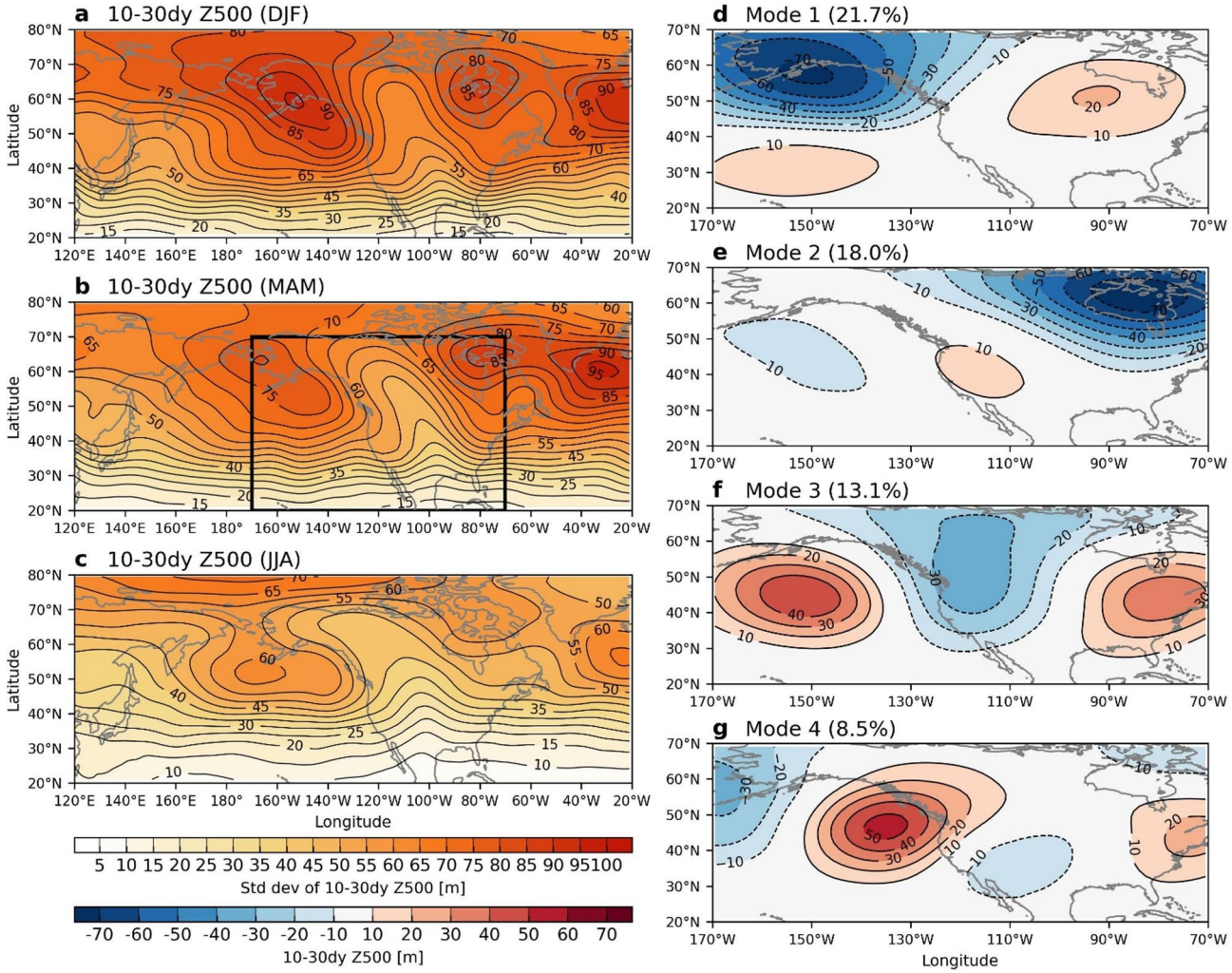


Fig. 1. Subseasonal modes in North Pacific-America regions during spring. Standard deviation of 10–30 bandpass filtered 500-hPa geopotential height (Z500; m) in **(a)** winter (December-January-February), **(b)** spring (March-April-May), and **(c)** summer (June-July-August) for the period of 1979–2023. In panels **(d)**-**(g)**, the four leading modes of empirical orthogonal function (EOF) analysis for springtime 10–30 day bandpass-filtered Z500 over eastern North Pacific-North America region (170°–70°W; 20°–70°N; see boxed region in **(b)**). The zero line is omitted. The explained variance by each EOF mode is denoted at the top in parenthesis.

Given the impacts of MJO on springtime North American climate[31–33], the relationship between MJO phases and subseasonal modes is further highlighted in Fig. S6. A notable relationship between the MJO and subseasonal modes is observed for Modes 1, 3, and 4, all of which have centers of action located in the North Pacific region, although their relationships are evident for only one phase of each mode. Specifically, MJO 1–2 phases lead the negative Mode 1 with a lag of 2–7 days (Fig. S6b). Mode 2, which is centered over northeastern North America, shows no clear relationship with MJO phases (Fig. S6c, d). For Mode 3, the positive phase, characterized by anticyclonic–cyclonic–anticyclonic circulation across the North Pacific-North America, demonstrates a strong connection with MJO phases 5–6, at lags − 4–0 days (Fig. S6f). Similarly, the negative phase of Mode 4 is linked with MJO phases 5–6 with a peak at lags from − 7 to −3 days of Mode 4's onset (Fig. S6h).

Modulation of springtime weather extremes by modes of subseasonal variability

How does the subseasonal variability delineated above modulate North American springtime extremes? To answer this question, we constructed composite maps of statistics quantifying surface extremes (Methods) for each of the four leading subseasonal modes. The probability distribution of PC values for each mode turns out to be largely symmetric about the mean (i.e., small skewness), suggesting a similar number of cases for each of the two opposite phases of a given mode (Fig. S7). In addition, since the overall composite results for the two phases largely mirror each other (Fig. S8), we focus our discussion on a single phase for each mode.

Figure 3 shows the statistical connections between surface weather extremes and the North American subseasonal variability represented by four leading modes. Mode 1 exhibits a pronounced cyclonic circulation anomaly over the North Pacific high latitude. Downstream of the circulation, significantly elevated frequencies of extremes (> 2 times) are found along the west coast of North America (Fig. 3a, b). The other high-latitude

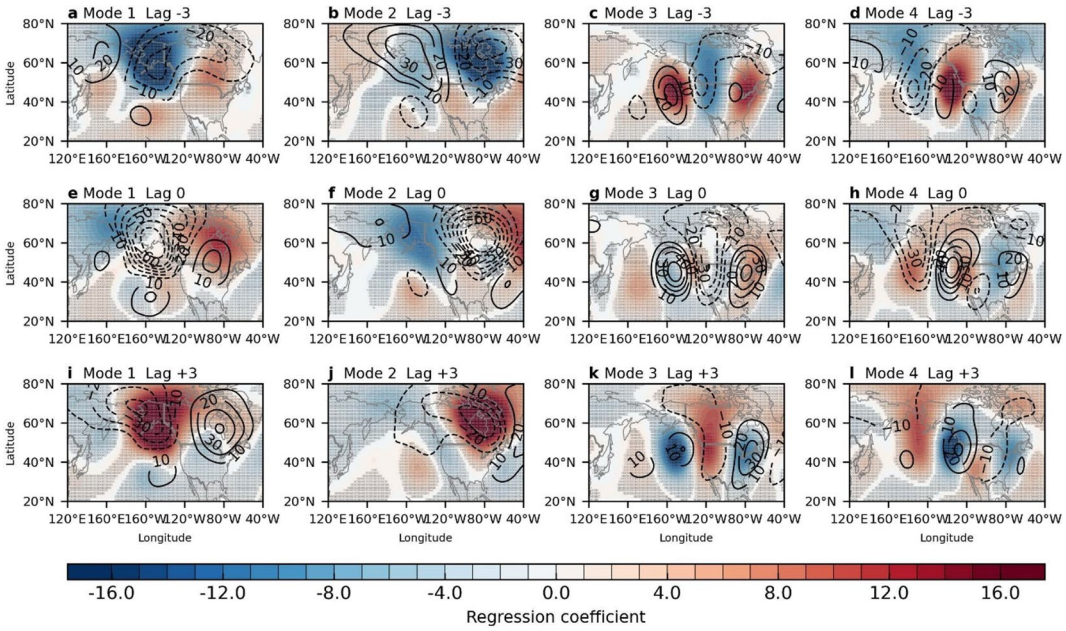


Fig. 2. Spatiotemporal evolution of subseasonal modes. Spatiotemporal evolution of four leading EOF modes from lag − 3 to 3 days of the onset. Contour and shading indicate the regression coefficient of Z500 (m) and its tendency (m day^{-1}) with respect to principal component (PC) time series of each EOF mode.

variability, Mode 2, shows a negative circulation anomaly covering the entire northeastern North America, accompanied by a weak anticyclonic circulation over the western United States. This mode is associated with an east-west dipole pattern in frequency anomalies of precipitation extremes across the United States (Fig. 3c): a decrease in precipitation extremes over the western United States and an increase in the east. The impact on wind extremes has a similar spatial distribution, exhibiting elevated frequencies in the northeastern United States and a weak drop of frequencies in the southwestern United States (Fig. 3d). Mode 3, characterized by the anticyclonic-cyclonic-anticyclonic Rossby wave packet extending from the eastern North Pacific to eastern North America, increases extreme precipitation frequencies (by up to ~ 60%) in the Southwest and Midwest of the United States (Fig. 3e). Wind extremes also occur more frequently (> 2 times) over the western and central United States especially over the Southwest, downstream of the cyclonic circulation (Fig. 3f). This suggests that variability in the tropical central Pacific may generate its footprints in North American springtime extremes through the excitation of subseasonal variability[64–67]. In Mode 4, the anticyclonic circulation anomaly reduces the frequency of precipitation extremes over western North America except for the southeastern Alaska (Fig. 3g). The accompanied cyclonic circulation anomaly downstream increases precipitation extremes over the central United States by about 50%. The wind extreme response to Mode 4 is characterized by a tripole pattern with decreased frequencies over the western United States and increased frequencies in the northwestern North America and the south-central United States adjacent to the Gulf of Mexico (Fig. 3h). It is worth pointing out that these four modes are also associated with anomalous distributions in the frequency of dry spells over North America that are consistent with local changes in precipitation extremes (Fig. S9), indicating equally important impacts of subseasonal variability on both types of hydrological extremes (i.e., extreme wetness and dryness). Similar results are obtained from the quantile regression analysis, supporting the robustness of these findings (Fig. S10,S11).

Coupling among subseasonal, synoptic- and mesoscale variability

To understand how subseasonal modes affect large-amplitude weather disturbances and consequently control the frequency of hydrological and wind extremes, the responses of the local synoptic- and mesoscale variability (i.e., weather activity) to four modes, in terms of composites based upon the PCs of the subseasonal modes, are displayed in Fig. 4. In Mode 1, with a cyclonic circulation anomaly over the North Pacific, enhanced synoptic variability (~ 25% of its subseasonal variation; Fig. S12a) is observed on the equator side of the circulation (Fig. 4a), partly due to a southward migration of synoptic systems exiting the Pacific storm track and/or the local strengthening of synoptic transients associated with changes in subseasonal background shear[68] in the region. Downstream of the cyclonic circulation anomaly is a region characterized by significantly elevated synoptic variability (~ 20%), which is likely linked to cyclogenesis east of a subseasonal trough. Mesoscale variability is also amplified on the equatorward flank and downstream of the cyclonic circulation anomaly (~ 45%; Fig. S13a), consistent with the fact that synoptic forcing is crucial in creating favorable environments for the development of mesoscale systems such as MCSs and frontal squall lines (Fig. 4b). Such a concurrent increase in synoptic- and mesoscale variability explains a substantial increase of precipitation and wind extremes over the west coast of North America. In the opposite phase of Mode 1, the responses of synoptic- and mesoscale variability in general

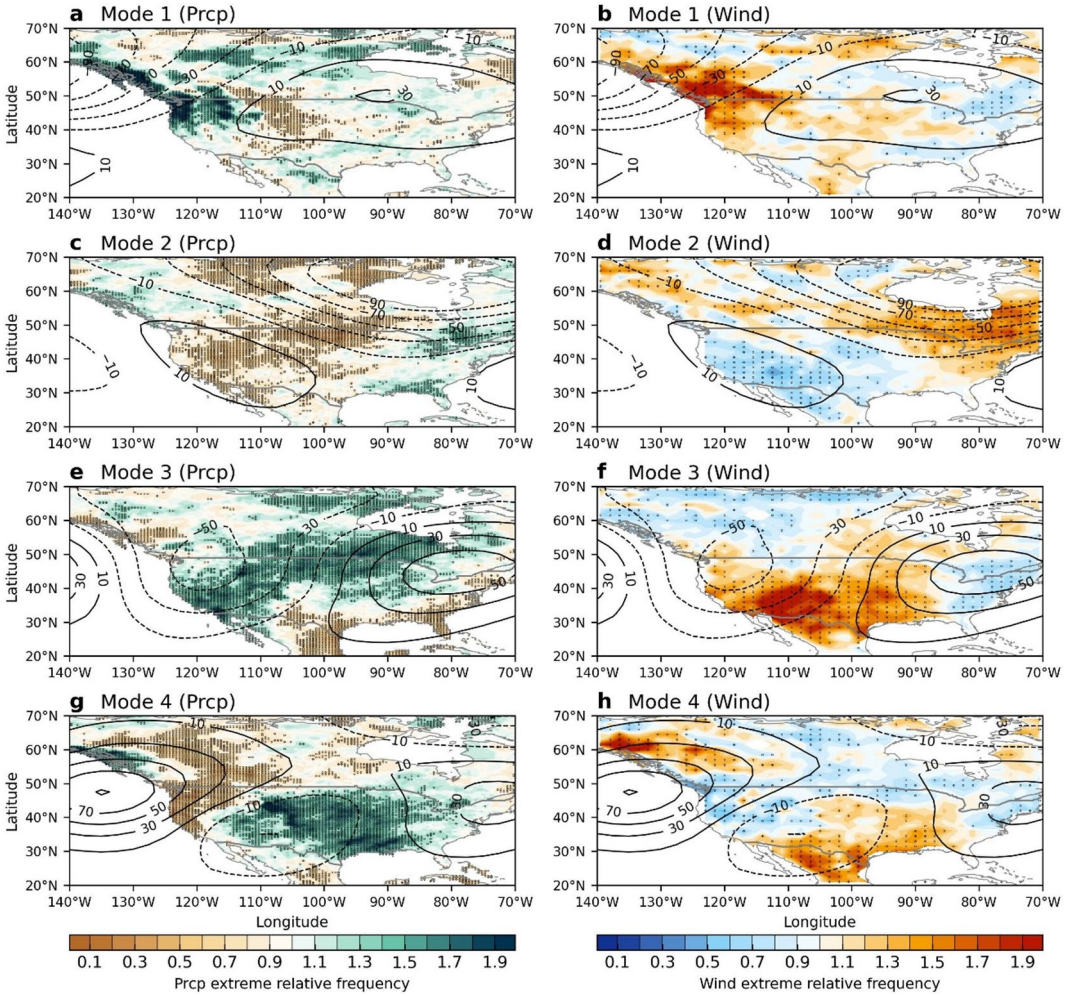


Fig. 3. Impact of subseasonal modes on North American springtime extremes. Composite of the frequency anomaly of precipitation (left, shading) and (right, shading) wind extremes associated with four EOF leading modes. The 10–30 day bandpass filtered Z500 (contour; 10 m interval) corresponding to each EOF mode is superimposed omitting the zero line. The EOF modes are defined when the magnitude of PC time series exceeds its ± 1 standard deviation with the largest magnitude among 1–4 PC values (Methods). Statistically significant tendencies at the 95% confidence level are dotted (2,000 bootstrap samples). Here, only one phase of each EOF mode is considered. The geographical boundary of the United States is marked.

show similar spatial distributions but with opposite signs (Fig. S14a, b). This is also true for the remaining three modes to be discussed (Fig. S14c-h; see also Fig. S12b, d, f, h and S13b, d, f, h).

Mode 2 projects enhanced synoptic variability (~ 35%; Fig. S12c) on the equator side of its cyclonic circulation anomaly, accounting for corresponding changes in wind and precipitation extremes (Fig. 4c). This is likely caused by the fact that the subseasonal high-latitude circulation anomaly modifies the preferred pathways of extratropical cyclones in the United States, including lee cyclones developing east of the Rockies[69,70]. Mesoscale variability, which is typically embedded within frontal systems of synoptic eddies[22], also shows a qualitatively similar response (~ 45%; Fig. S13c), facilitating the corresponding surface extreme changes (Fig. 4d). In southwestern United States, reduced synoptic- and mesoscale variability along the anticyclonic circulation anomaly is observed, consistent with a decreased frequency in precipitation and wind extremes.

In Mode 3, the cyclonic circulation over western North America as a part of the wave train strengthens both synoptic and mesoscale variability (~ 50%; Fig. S12e,S13e) over its western flank through modulating the trajectories of North Pacific storm track eddies[68] subsequently increasing the wind and precipitation extremes in this region. The south-central United States experiences enhanced synoptic activity related to elevated cyclogenesis activity downstream of the subseasonal trough (Fig. 4e). Strengthened synoptic variability in turn plays a key role in the formation and maintenance of MCSs in the region through providing both lifting for convection initiation and part of the low-level poleward moisture transport (Fig. 4f; see also Fig. S15a)[28,29,71–73], ultimately increasing wind extremes in the Great Plains. The subseasonal anticyclone over the eastern North America deflects extratropical cyclones northward and enhances poleward moisture transport along its western flank, which explains in a similar way the increased precipitation and wind extremes over the northern United States and eastern Canada.

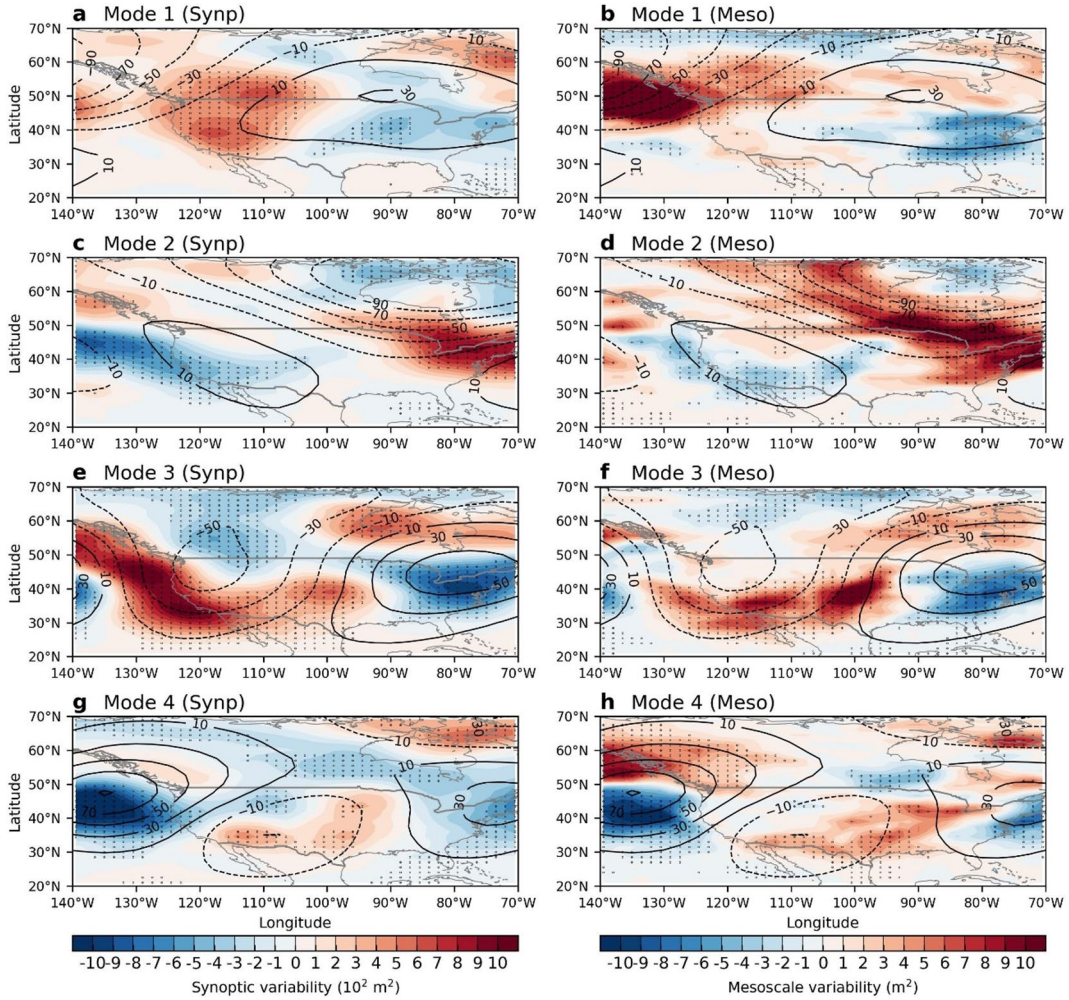


Fig. 4. Changes in synoptic- and mesoscale variability associated with subseasonal modes. Same as Fig. 3 but for the anomaly of synoptic variability (left, Unit: 10^2 m^2) and mesoscale variability (right, Unit: m^2).

The anticyclonic circulation anomaly of Mode 4 deflects synoptic eddies from downstream of the Pacific storm track and leads to amplified synoptic- and mesoscale variability along its northern flank (Fig. 4g, h). This is responsible for the increased wind and precipitation extremes over the coastal region of southeastern Alaska. Downstream of Mode 4's cyclonic circulation anomaly over the southwestern United States, both synoptic- and mesoscale variability are significantly enhanced (15 ~ 30% increase; Fig. S12g,S13g). It reflects again the enhanced MCS activity (Fig. S15b), accounting for the increased precipitation and wind extremes over the central United States. Note that despite the moderate magnitudes of the variability amplifications, the increase of precipitation extremes in this region is substantial (Fig. 3g), likely associated with the abundance of moisture over the Gulf States.

To understand large-scale factors, linked to physical relationships across different scale variabilities, changes in pressure velocity and moist transport are analyzed[24,28,71–73]. A 10–30 day bandpass filter is applied to the vertical motions, focusing on large-scale divergence (or convergence) in upper-level circulations through mass continuity (Fig. 5b, e, h, k). For moisture transport, integrated water vapor transport and low-tropospheric moisture flux are examined (Fig. 5a, d, g, j). It is noted that while these variables can independently create favorable conditions for enhanced precipitation, they also modulate the activity of synoptic and mesoscale variability[15,16]. Additionally, we evaluate changes in frontogenesis modulated by subseasonal modes, a process known to significantly influence the United States precipitation patterns[19]. It bridges, at least in part, the connection between synoptic and mesoscale variability[22] (Fig. 5c, f, i, l).

In Mode 1, the cyclonic circulation over the North Pacific facilitates substantial moisture transport from the ocean, creating favorable conditions for enhanced synoptic- and mesoscale variability that in turn contribute to precipitation and wind extremes in west coast regions (Fig. 5a). Strong ascending motions attributed to upper-level divergence and enhanced frontogenesis, coupled with orographic effects[74], further support the increased precipitation extremes along the west coast region (Fig. 5b, c). The anticyclonic height anomaly over the southwestern United States in Mode 2 reduces northward moisture transport from the Gulf in these regions (Fig. 5d). The cyclonic circulation anomaly of Mode 2 assists to decrease northward moisture transport along its southwestern flank with a downward motion (Fig. 5e) and weakened frontogenesis (Fig. 5f). These changes

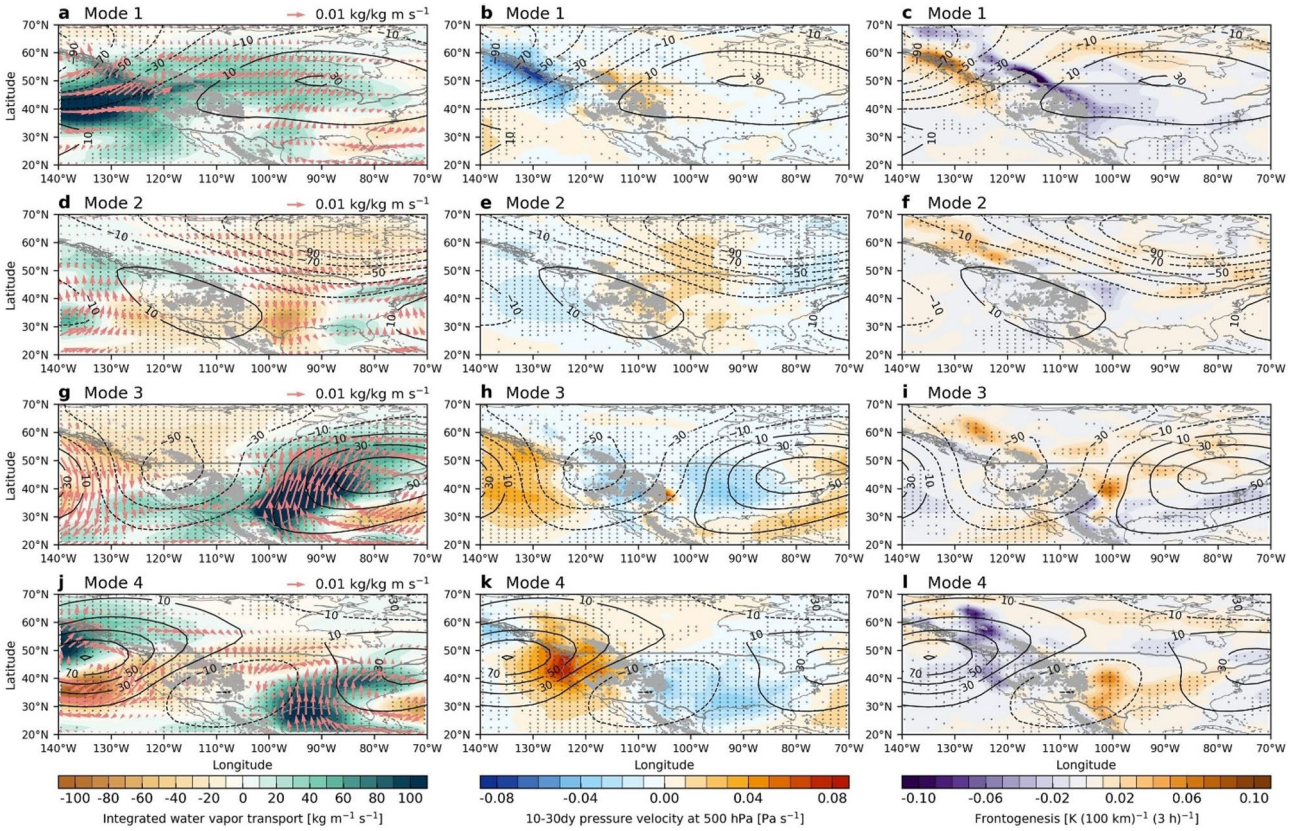


Fig. 5. Dynamical and thermodynamic factors associated with synoptic- and mesoscale variabilities and their corresponding extremes. Same as Fig. 3 but for (a, d, g, j) integrated water vapor transport anomaly (shading; kg m^{-1} s^{-1}) and 850-hPa moisture flux anomaly (vector; kg kg^{-1} m s^{-1}), (b, e, h, k) 10–30 day bandpass filtered pressure velocity at 500 hPa (shading; Pa s^{-1}), and (c, f, i, l) frontogenesis anomaly (shading; K (100 km)$^{-1}$ (3 h)$^{-1}$). The gray shading denotes the regions with a surface elevation higher than 2500 m. For visual purposes, a nine-point filter (weighted sum) is applied to pressure velocity and frontogenesis anomaly after calculation.

subsequently decrease precipitation extremes, coupled with reduced synoptic and mesoscale variability. In its southeast flank, although weak, enhanced moisture transport and upward motion create favorable conditions for synoptic and mesoscale variability. It also explains the frequent surface wind and precipitation extremes in the northeastern United States. In Mode 3, cyclonic and anticyclonic circulations embedded in a Rossby wave train drive a strong northward moisture transport (Fig. 5g). In the region between two circulation anomalies, enhanced ascending motions due to upper-level divergence (Fig. 5h) and frontogenesis (Fig. 5i) are observed, accounting for concurrent changes in synoptic and mesoscale variability over North America. Similarly, the anticyclonic circulation of Mode 4 generates a dipole pattern of moisture transport anomalies in western North America (Fig. 5j). The enhanced mesoscale variability with strong moisture transport over the northern flank of circulation explains the localized increase in wind and precipitation extremes on the west coast of Canada. Along with the moisture transport change, pronounced downward motions with upper-level convergence and weakened frontogenesis are observed over the southern flank of the anticyclonic circulation. These changes align with decreased synoptic- and mesoscale variability (Fig. 5k, l). Its downstream, cyclonic circulation anomaly also facilitates a strong moisture transport into the south-central United States. The pronounced ascending motions and frontogenesis explain enhanced synoptic- and mesoscale variability as well as increased precipitation extremes in the region. This result suggests that the changes in the dynamical and thermodynamic large-scale environments associated with subseasonal modes demonstrate consistency with the corresponding changes found in both synoptic- and mesoscale variabilities and their associated surface extremes. The opposite phases of subseasonal modes also show qualitatively similar responses with reversed signs (Fig. S16).

Decadal changes of subseasonal variability and the associated footprints in weather extremes

The above results demonstrate that springtime subseasonal variability shapes the synoptic- and mesoscale weather disturbances over North America and ultimately modulates the activity of weather extremes. It suggests that their changes can be linked with the North American extreme variability on a longer timescale. Figure 6 illustrates the long-term evolution of the activity of the four subseasonal modes in terms of the spring anomalous (relative to its climatology) probability density functions (PDFs) of their corresponding PCs. Distinct decadal-

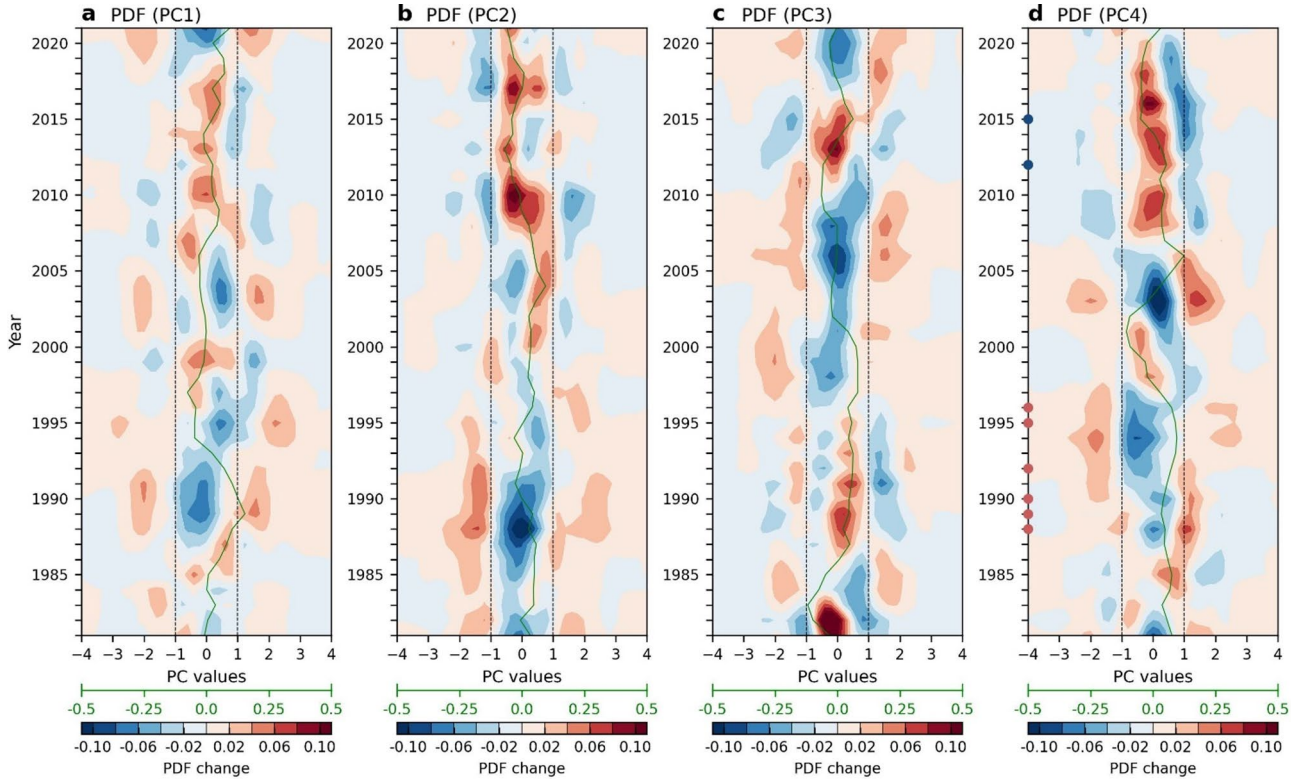


Fig. 6. Long term changes in PDF of subseasonal modes. Probability density function changes in PC time series compared to its climatology within 5-year window for four leading EOF modes. The median of PC values within 5-year window at given year is denoted as green solid line. As a reference, ± 1 standard deviation line of PC value is superimposed (dashed line). In Panel d, hydrological extreme years over central US regions (110°–90°W; 30°–45°N) are highlighted as red and blue dots during active (1988–1997) and muted (2008–2017) periods of EOF mode 4, respectively.

scale changes are found in the degree of the general "activity" of the four modes with "active" periods indicated by positive PDF anomalies at the tails corresponding to distributions with high kurtosis and "muted" periods indicated by negative PDF anomalies at the tails and positive anomalies around the mean corresponding to distributions with low kurtosis. Specifically, Mode 1 exhibits an active period in the 1990s (Fig. 6a) and is characterized in general by pronounced interannual variations in its activity. Mode 2 is particularly active in the 1980s and 1990s followed by a muted period in the first two decades of the 21st century (Fig. 6b). The decadal-scale changes in Mode 3 are characterized by an active period from the mid-1990s to the 2010s (Fig. 6c), whereas Mode 4 reveals both active and muted decades in 1990s and 2010s (Fig. 6d). For regions that are mostly affected by a single subseasonal mode, the impact due to mutual interferences of multiple modes in any individual year is minimized, and this allows a potential direct projection of the decadal-scale changes of the activity of subseasonal modes onto the decadal-scale changes in the activity of weather extremes. The south-central United States (110°–90°W, 30°–45°N) is a good example, where hydrological extremes are largely modulated by Mode 4 in boreal spring (Fig. 3g). If a year of elevated frequency of hydrological extremes is identified when the total number of grid points in the region showing springtime precipitation extremes or dry spells exceeds its year-to-year standard deviation, six out of ten years during Mode 4's active decade (1988–1997) are characterized by elevated frequency of hydrological extremes (red dots in Fig. 6d). In contrast, only two extreme years (blue dots) are found during the decade of muted Mode 4 activity (2008–2017). This result suggests that for specific regions subseasonal variability may play a crucial role in changing the occurrence frequency of weather extremes across decadal and even longer timescales.[68–73]

Discussion

In light of these findings, we propose and outline here a "scale ladder" climbing and descending, which allows one to effectively dissect the multi-layered dynamics of regional extremes and potentially understand causes of long-term changes in these extremes. As sketched in Fig. 7, modes of subseasonal variability act as a critical step in this scale ladder that bridges potential natural and anthropogenic forcings with variations in regional weather extremes. More specifically, atmospheric subseasonal variability is modulated by both interannual to decadal variability internal to the climate system and long-term anthropogenic changes in the boundary conditions (e.g., sea surface temperatures) through direct excitation of quasi-stationary waves and/or changes in atmospheric mean state thus instability properties[75–80]. For instance, Figure S17 shows the detrended interannual time series of the Pacific Decadal Oscillation (PDO) index and the activity of the four subseasonal modes, represented by

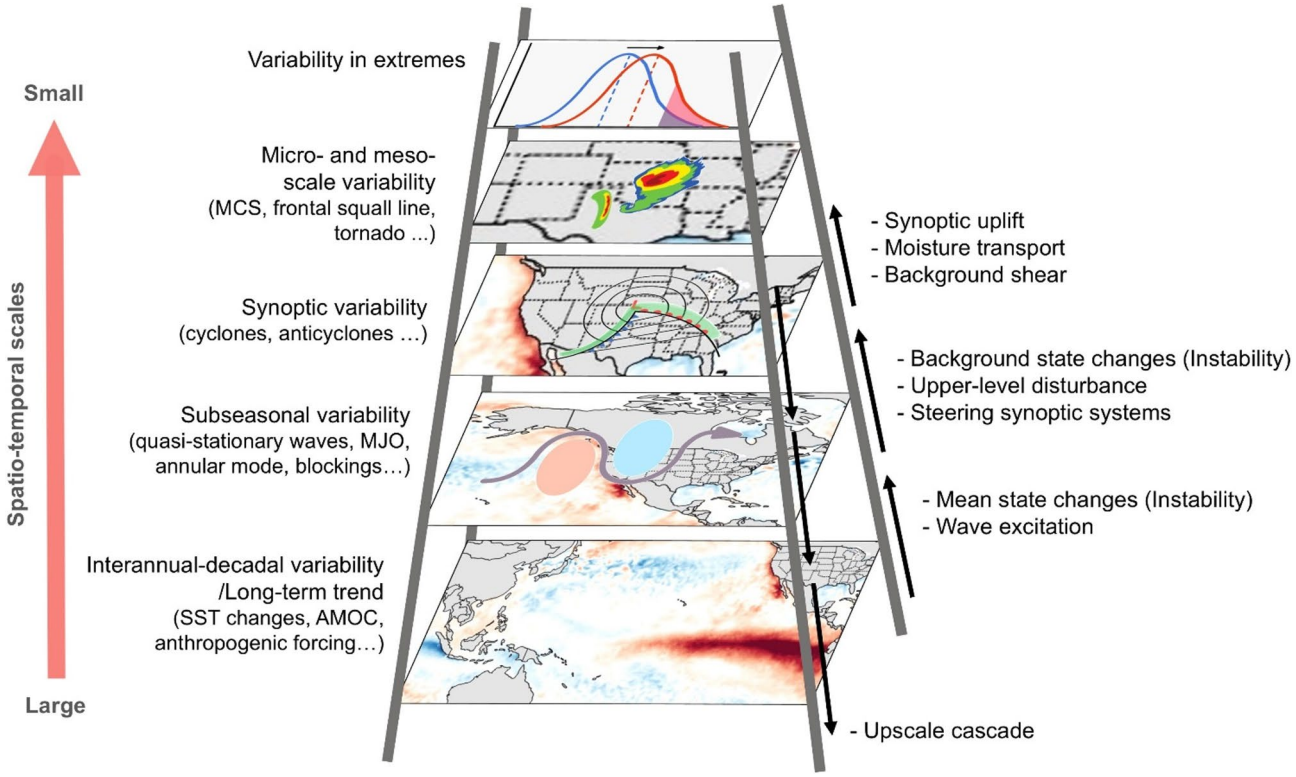


Fig. 7. Schematic diagram of scale ladder associated with North American springtime extremes. The ladder rungs represent the variability from interannual-decadal to micro-mesoscale (bottom-up). The upward and downward arrows indicate potential mechanisms between large-scale and small-scale variability at each rung. The top of the ladder shows changes in extreme variability systematically affected by multi-scale processes, ranging from interannual-decadal to micro-mesoscale.

the standard deviation of the PC values for a given season. For Modes 2 and 4, their interannual correlations with the PDO index (r: 0.17–0.18) are higher compared to the case when the linear trends are removed (r: 0.002–0.05) (Fig. S17b, d). This suggests that both the PDO index and these two modes have a shared long-term trend, possibly linked to anthropogenic forcing. In contrast, Mode 3 exhibits a significant interannual relationship with the PDO (r: −0.29 without trend, −0.34 with trend; Fig. S17c), confirming its robust connection with the atmosphere-ocean coupled variability over the North Pacific. The findings regarding the long-term variability of Mode 3 are also partly in line with those of previous studies that examined the relationship between the MJO and long-term variability[81–84], due to the impact of the MJO on Mode 3 discussed earlier.

Going back to the scale ladder, modes of subseasonal variability[85–88] modify extratropical circulation (e.g., jet location and intensity), thereby affecting the local energetics[89,90] and trajectories of regional synoptic-scale weather systems[91]. The resulting changes in synoptic variability further translate into changes in synoptic forcing (e.g., uplifting, moisture transport, and wind shear) that either enhance or suppress activities of meso- and micro-scale weather disturbances. Variations in meso- and micro-scale variability are ultimately linked with changes in the frequency and/or intensities of weather extremes such as tornadoes, hailstorms, and extreme rainfall. This schematic further reveals that information flows in both directions between steps of this scale ladder. For instance, synoptic-scale disturbances provide feedbacks to their subseasonal background through transports of heat, momentum, and moisture, and convective heating associated with mesoscale weather disturbances may effectively modify the growth rate and scale of synoptic disturbances such as extratropical cyclones[92,93]. In other words, the cascade of energy is often bi-directional between steps of the ladder even though on many occasions one direction appears more prominent.

The "scale ladder" introduced here is based upon the fact that the spectrum of atmospheric motion is continuous and covers a wide range. We adopt this terminology to emphasize the importance of recognizing the multi-scale nature of the atmosphere in order to obtain mechanistic understanding of weather extremes that typically occur at the lower end of the spectrum. "Ladder" was used particularly to highlight the great amount of effort and labor it often takes for one to climb up and down a ladder to achieve this goal. Such a multi-scale perspective is obviously not new. For instance, part of our results, the relationship between subseasonal and longer-timescale variability, can be partly linked with the Fluctuation-Dissipation theorem[94] and similar ideas explored in previous studies[95,96]. The relationship between subseasonal and smaller-scale (i.e., synoptic- to mesoscale) variability discussed is also in line with those obtained through multi-scale analysis approaches[29,91]. The novel aspect of our study lies in the integration of the known ideas of scale interaction into a perspective and method of "scale ladder" that can, and in our humble opinion, should be applied systematically to address

various questions related to weather extremes including dynamical origin and long-term changes. The idea is not restricted to any individual season nor any specific geographic location, even though our analysis initially focused on North America in boreal spring. Climbing and descending this ladder after characterizing first the scale of disturbances directly responsible for the formation of a weather extreme allows one to chart a clean dynamical pathway that traces an extreme event's occurrence back to its origin. As a result of unfolding multi-layered dynamics of extremes, such a pathway facilitates the explanation of dynamically driven variability and changes in extremes across various timescales, for both those internal to the climate system and those externally forced. In a similar fashion, this scale ladder can be used to effectively identify sources of predictability for extremes. Applying the ladder to model simulated extremes and contrasting the associated model pathways with the observed ones also provide a powerful tool for understanding and tracking the origins of potential model biases in representing extremes especially their variability and long-term changes. It is noted that the physical and dynamical processes bonding adjacent steps of the scale ladder, some of which are well recognized and studied, still need to be systematically evaluated and quantified such that the ladder can be customized for a given region and/or a specific type of extreme. The dynamical nature of the steps representing subseasonal variability needs further clarification, and the capability of the multi-scale perspective highlighted here in explaining regional changes of extremes in a warming climate has yet to be examined. We also want to stress here that this diagnostic analysis is based upon a single reanalysis product. Uncertainties remain in the methods and threshold values used for quantifying extremes as well as in the estimated amplitudes of variabilities, especially those of mesoscales. All of these demand significant research efforts and will be in our future work.

Methods

Data

The hourly and 6-hourly data from the 5th generation reanalysis of the European Centre for Medium-Range Weather Forecasts (ERA5)[97] is utilized for all the diagnoses concerning the boreal springs (March-April-May) in the period of 1979–2023. Global gauge-based daily precipitation data from the Climate Prediction Center (CPC)[98] is adopted to identify and assess hydrological extremes. All variables, except for precipitation (0.5° by 0.5°), are set to 1.5° by 1.5° horizontal resolution, and their daily values are obtained by taking an average over the four 6-hourly values (i.e., 00, 06, 12, 18 UTC).

Subseasonal variability

Subseasonal variability over North America is identified by performing an empirical orthogonal function (EOF) analysis for 10–30 day bandpass-filtered geopotential height at 500 hPa (Z500) over the eastern North Pacific to North America sector (170°–70°W, 20°–70°N). The four leading North American subseasonal modes, explaining 60% of total variability, are examined in this study. To investigate the impacts of each EOF mode on surface extremes and the corresponding projections onto synoptic- and mesoscale variability, we performed a composite analysis for each EOF mode utilizing PC values exceeding ± 1 standard deviation to construct opposite phases. If a given day satisfies the threshold in multiple EOF modes, we assigned it to the case of EOF mode showing the largest PC value.

Synoptic- and mesoscale variability

The activity of synoptic-scale variability is quantified in terms of the square of the 2–10-day bandpass filtered daily Z500 field[52]. Similarly, mesoscale variability is quantified by the square of the 24-hour high-pass filtered hourly Z500 with diurnal cycle removed (i.e., hourly-mean; 24 timesteps). This timescale captures the sub-daily variability associated with both convective activity (e.g., MCS over the Great Plains) and passages of fronts within synoptic systems.

Hydrological and surface wind extremes

This study examines both hydrological (i.e., wet and dry) and surface wind extremes over North America. A precipitation extreme at a grid point is identified when daily precipitation anomaly (i.e., departure from the calendar day climatology) exceeds its 90th percentile. A dry extreme is detected when daily precipitation is less than 1 mm per day[99] for 10 consecutive 10 days[100]. A wind extreme at a grid point is identified if the daily maximum surface (10 m) wind speed anomaly exceeds its 90th percentile. The daily maximum wind speed is the highest hourly wind speed over a 24-hour span in ERA5. The impacts of the leading modes of subseasonal variability on extremes are quantified at a grid point by the ratio between the occurrence probability of the extremes at different phases of each subseasonal mode and the climatological occurrence probability of extremes[101,102].

Moisture transport

To understand the moisture transport changes associated with subseasonal variability, moisture flux anomaly at 850 hPa and vertically integrated water vapor transport (IVT) are examined. Here, IVT is computed by integrating zonal and meridional moisture flux from 1000 hPa to 300 hPa:

$$\mathrm{IVT} = \left[\left(-\frac{1}{g} \int_{1000\,hPa}^{300\,hPa} qu dp \right)^2 + \left(-\frac{1}{g} \int_{1000\,hPa}^{300\,hPa} qv dp \right)^2 \right]^{1/2} \quad (1)$$

where g is the gravitational acceleration (9.806 m s^{-2}); q is the specific humidity (kg kg^{-1}); u and v are the zonal and meridional winds (m s^{-1}), respectively.

Blocking index

Atmospheric blocking is identified using a Lagrangian blocking index based on 500-hPa geopotential height anomaly fields[41,42,103]. The large-scale (> 2.5 × 10^6 km^2 spatial scale) positive height anomaly, defined as a departure from its daily climatology, is detected when it exceeds the monthly-varying threshold. Then, a blocking event is finally identified when the detected anomaly shares day-to-day overlapping areas (> 0.5 ratio) for at least 5 consecutive days. Here, the monthly-varying threshold is defined as 1.3 standard deviations of 500-hPa geopotential height anomaly distribution. The distribution is calculated across all grid points located north of 30°N and all days within a 3-month centered window.

Frontogenesis

Frontogenesis (F) is assessed using Petterssen's two-dimensional kinematic frontogenesis function[104], based on potential temperature at 850 hPa:

$$F = \frac{1}{2} |\nabla \theta| (Dcos2\beta - \delta) \tag{2}$$

where D is the total deformation and β is the local angle between the dilatation axis and isentropes. δ is the horizontal divergence; θ is the potential temperature.

Data availability

The data used in this study are available at https://cds.climate.copernicus.eu/datasets (ERA5) and at https://downloads.psl.noaa.gov/Datasets/cpc_global_precip (CPC precipitation). The MCS tracking data are available at https://zenodo.org/records/10023498, and the PDO index was obtained from the NOAA NCEI (https://www.ncei.noaa.gov/access/monitoring/pdo/).

Code availability

The analysis was carried out using the function in NCAR Command Language v.6.2.1, Python packages (v3.11.5), and R (v4.2.0). To carry out the interpolation of the model grid data, we utilized climate data operators available at https://code.mpimet.mpg.de/projects/cdo. All figures are generated using Python with Cartopy for map projections.

Received: 31 May 2025; Accepted: 29 September 2025
Published online: 05 November 2025

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Acknowledgements

We are grateful to three anonymous reviewers for their constructive suggestions. J.H., Z.Y., and Y.D. are supported by the U.S. National Science Foundation (NSF) through Grant AGS-2032532 and by the U.S. National Oceanic and Atmospheric Administration (NOAA) through Grant NA22OAR4310606.

Author contributions

J.H. conducted the analysis, illustrated the figures, and wrote the manuscript. Y.D. conceptualized the overall research idea and wrote the manuscript. Z.Y. and H.K. improved the interpretation of the results and helped to write the manuscript. All of the authors discussed the results and reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-22366-8.

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